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Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

**Course Code: MAT201****Course Name: Partial Differential equations and Complex analysis**

Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions. Each question carries 3 marks*

Marks

- 1 Derive a partial differential equation from the relation  $z = (x + y) f(x^2 - y^2)$  (3)
- 2 Solve using direct integration  $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$  (3)
- 3 Solve  $2z = xp + yq$ . (3)
- 4 Write any three assumptions in deriving one dimensional heat equation. (3)
- 5 Show that an analytic function  $f(z) = u + iv$  is constant if its real part is constant. (3)
- 6 Show that the function  $u = \sin x \cos hy$  is harmonic. (3)
- 7 Find the Maclaurin series of  $f(z) = \sin z$  (3)
- 8 Evaluate  $\oint_C \ln z \, dz$ , where  $C$  is the unit circle  $|z| = 1$ . (3)
- 9 Find all singular points and residue of the function  $\operatorname{cosec} z$  (3)
- 10 Determine the location and order of zeros of the function  $\sin^4\left(\frac{z}{2}\right)$  (3)

**PART B***Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11 (a) Form the Partial differential equation by eliminating the arbitrary constants from  $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$  (5)
- (b) Solve  $2xz - p x^2 - 2qxy + pq = 0$  (9)
- 12 (a) Solve  $\frac{\partial^3 z}{\partial^2 x \partial y} = \cos(2x + 3y)$  (7)
- (b) Solve  $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$  (7)

**Module 2**

- 13 (a) Derive the solution of the one dimensional wave equation  $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$  using variable separable method. (6)
- (b) An insulated rod of length  $l$  has its ends A and B maintained at  $0^\circ \text{C}$  and

$100^{\circ}\text{C}$  respectively until steady state conditions prevail. If B is suddenly reduced to  $0^{\circ}\text{C}$  and maintained at  $0^{\circ}\text{C}$ , find the temperature at a distance  $x$  from A at time  $t$ . (8)

14 (a) Derive the one dimensional heat flow equation. (6)

(b) A tightly stretched string of length  $l$  with fixed ends is initially in equilibrium position. If it is set vibrating by giving each points a velocity  $v_0 \sin^3(\frac{\pi x}{l})$ . Find the displacement  $y(x,t)$ . (8)

### Module 3

15 (a) Find an analytic function whose real part is  $u = \sin x \cosh y$  (7)

(b) Find the image of the strip  $\frac{1}{2} \leq x \leq 1$  under the transformation  $w = z^2$  (7)

16 (a) Check whether  $w = \log z$  is analytic. (8)

(b) Show that under the transformation  $w = \frac{1}{z}$ , the circle  $x^2 + y^2 - 6x = 0$  is transformed into a straight line in the  $W$  plane. (6)

### Module 4

17 (a) Integrate counter clockwise around the unit circle  $\oint_C \frac{\sin 2z}{z^4} dz$  (7)

(b) Find the Taylor series of  $\frac{1}{1+z}$  about the centre  $z_0 = i$ . (7)

18 (a) Evaluate  $\int_0^{1+i} (x - y + ix^2) dz$  along the parabola  $y = x^2$ . (7)

(b) Evaluate  $\oint_C \frac{\log z}{(z-4)^2} dz$  counter clockwise around the circle  $|z - 3| = 2$ . (7)

### Module 5

19 (a) Find the Laurent's series expansion of  $\frac{z^2 - 1}{z^2 - 5z + 6}$  about  $z = 0$  in the region  $2 < |z| < 3$  (5)

(b) Evaluate  $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta}$ . (9)

20 (a) Evaluate  $\oint_C \frac{z-23}{z^2 - 4z - 5} dz$  where  $C : |z - 2 - i| = 3.2$  using Residue theorem. (5)

(b) Evaluate  $\int_0^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)}$ . (9)

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## APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

### Scheme for Valuation/Answer Key

*Scheme of evaluation (marks in brackets) and answers of problems/key*  
**Third Semester B.Tech Degree Examination December 2020 (2019 scheme)**

**Course Code: MAT201**

**Course Name: Partial Differential Equations and Complex analysis**

Duration: 3 Hours

### PART A

*Answer all questions. Each question carries 3 marks*

Marks

- |   |                                                                                                                                                                                                                                                                                                                                  |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1 | $z = (x+y) f(x^2 - y^2)$ .<br>Finding p and q (1).<br>Eliminating $f(x^2 - y^2)$ (1).<br>Forming the PDE ( $py + qx = z$ ) (1).                                                                                                                                                                                                  | (3) |
| 2 | $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$ .<br>Integrating partially with respect to x<br>$\frac{\partial u}{\partial t} = e^{-t} \sin x + f(t)$ (1).<br>Integrating partially with respect to t<br>$u = -e^{-t} \sin x + \int f(t) dt + g(x)$ . (2)                                                          | (3) |
| 3 | Forming the subsidiary equations<br>$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{2z}$ (1).<br>Solving the subsidiary equations and finding two sets of solutions<br>(1)<br>$\frac{x}{y} = a$<br>$\frac{y^2}{z} = b$ .<br>Writing the solution of the equation $2z = xp + yq$ as<br>(1)<br>$f\left(\frac{x}{y}, \frac{y^2}{z}\right)$ | (3) |



(1).

(Marks shall be given to any alternate solution)

4 Assumption 1 (1) (3)

Assumption 2 (1)

Assumption 3 (1)

5  $u$  is constant, then  $\frac{\partial u}{\partial x} = 0$ ,  $\frac{\partial u}{\partial y} = 0$  (3)

(1).

By CR equations  $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 0$  and  $\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 0$ .

(1).

Therefore  $v$  is a constant. Hence  $f(z)$  is a constant.

(1)

6  $u = \sin x \cosh y$ . (3)

$$\frac{\partial u}{\partial x} = \cos x \cosh y.$$

$$\frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y. \quad (1)$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y.$$

$$\frac{\partial^2 u}{\partial y^2} = \sin x \cosh y. \quad (1)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

 $U$  satisfies Laplace equation and hence harmonic. (1)7  $f(z) = \sin z$ . (3)Finding  $f^1(z)$ ,  $f^{11}(z)$ ,  $f^{111}(z)$ , ..Finding  $f^1(0)$ ,  $f^{11}(0)$ ,  $f^{111}(0)$ , .. (1)

Substituting in

$$f(z) = f(0) + \frac{z}{1!} f^1(0) + \frac{z^2}{2!} f^{11}(0) + \frac{z^3}{3!} f^{111}(0) + \dots \quad (1)$$

$$\sin z = \frac{z}{1!} - \frac{z^3}{3!} + \dots \quad (1)$$

8  $|z|=1$ ,  $z = e^{i\theta}$   $dz = ie^{i\theta} d\theta$  (1) (3)

Evaluating the integral using the above substitution



$$\oint_C \ln z \, dz = 2\pi i \quad (2)$$

$$9 \quad f(z) = \operatorname{cosec} z = \frac{1}{\sin z} . \quad (3)$$

$$= \frac{1}{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots} . \quad (1)$$

$$= \frac{1}{z} \left( 1 - \left( \frac{z^2}{3!} + \frac{z^4}{5!} - \dots \right) \right)^{-1} .$$

$$= \frac{1}{z} + \frac{z}{3!} - \dots \quad (1)$$

$z=0$  is a pole.

Residue at  $z=0$  is 1. (1)

$$10 \quad f(z) = \sin^4 \left( \frac{z}{2} \right). \quad (3)$$

$$f(z) = 0 \text{ for } z = 0, \pm 2\pi, \pm 4\pi, \dots \quad (1)$$

$$f'(z), f''(z), f'''(z) = 0 \text{ and } f^{(4)}(z) \neq 0 \text{ at these points.} \quad (1)$$

So the zeros are of the fourth order. (1)

(3)

### PART B

*Answer any one full question from each module. Each question carries 14 marks*

#### Module 1

$$11(a) \quad (x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha \quad (5)$$

Finding p and q (3).

$$p^2 + q^2 = \tan^2 \alpha \quad (2)$$

$$(b) \quad f(x,y,z,p,q) = 2xz - p x^2 - 2qxy + pq . \quad (1)$$

writing Charpits subsidiary equations (2) (9)

finding p and q from the subsidiary equations (3).

Substituting p and q in  $dz = pdx + qdy$  (2).

Solving and finding  $z = ay + b(x^2 - a)$  (1)

$$12(a) \quad \text{Integrating Partially with respect to } x \text{ (2 times) , Integrating partially} \quad (7)$$

with respect to y (3)

Writing the solution .

$$Z = \frac{-\sin(2x+3y)}{12} + x v(y) + u(y) + w(x). \quad (3)$$



Where  $v(y)$ ,  $u(y)$  are functions of  $y$  only  
and  $w(x)$  is a function of  $x$  only. (1)

(b)

$$x^2 (y - z)p + y^2 (z - x)q = z^2 (x - y).$$

Writing the subsidiary equations (1)

Forming two sets of solutions (4). (7)

Writing the solution

$$xyz = f\left(\frac{1}{x} + \frac{1}{y}, \frac{1}{z}\right) \quad (2).$$

Any other alternate solution will get full marks.

### Module 2

13(a) Derivation (6)

(b) The equation of Heat conduction  $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$  (1)

$$\text{Initial condition } u(x, 0) = \frac{100x}{l} \quad (1) \quad (8)$$

Boundary conditions

$$u(0, t) = 0 \text{ for all } t, \quad u(l, t) = 0 \text{ for all } t \quad (1)$$

$$\text{Solution } u(x, t) = (c_1 \cos px + c_2 \sin px) e^{-c^2 p^2 t}. \quad (1)$$

Applying the conditions and writing the solution

$$U(x, t) = \frac{200}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi x}{l} e^{-\left(\frac{n\pi c}{l}\right)^2 t} \quad (4)$$

**\*\* Give marks for any fair attempt irrespective of l (small L) or 1 (one)**

14(a) Derivation. (6)

(b) The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad (1)$$

Writing boundary conditions and initial conditions. (2)

The solution is of the form

$$Y(x, t) = (c_1 \cos px + c_2 \sin px) (c_3 \cos cpt + c_4 \sin cpt). \quad (1)$$

Applying the conditions and simplifying



$$y(x,t) = \frac{3v_0 l}{4\pi c} \sin\left(\frac{\pi x}{l}\right) \sin\left(\frac{\pi ct}{l}\right) - \frac{v_0 l}{12\pi c} \sin\left(\frac{3\pi x}{l}\right) \sin\left(\frac{3\pi ct}{l}\right) \quad (4)$$

### Module 3

$$15(a) \quad U = \sin x \cosh y. \quad (7)$$

$$u_x = \cos x \cosh y \quad u_{xx} = -\sin x \cosh y$$

$$u_y = \sin x \sinh y \quad u_{yy} = \sin x \cosh y.$$

$$u_{xx} + u_{yy} = 0 \quad . U \text{ is harmonic.} \quad (2)$$

To find

$$V = \sin x \cosh y + c \quad (2)$$

$$\text{To find } f(z) = \sin z + ic \quad (3)$$

$$(b) \quad W = Z^2$$

$$u = x^2 - y^2, \quad v = 2xy \quad (1)$$

$$\text{when } x = \frac{1}{2} \quad v^2 = \frac{1}{4} - u, \text{ which is a parabola.} \quad (2). \quad (7)$$

$$\text{When } x = 1 \quad v^2 = 4(1 - u), \text{ which is a parabola} \quad (2).$$

Hence the image of the strip  $\frac{1}{2} \leq x \leq 1$  is the region between the parabolas

$$v^2 = \frac{1}{4} - u \text{ and } v^2 = 4(1 - u). \quad (2)$$

$$16(a) \quad W = \log z \quad (8)$$

$$= u + iv = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right) \quad (2)$$

$$\text{Finding } u_x, u_y, v_x, v_y \quad (2)$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad (2)$$

CR equations are and partial derivatives are continuous . Hence

$$F(z) \text{ is analytic} \quad (2)$$

$$(b) \quad W = \frac{1}{z}$$

$$x = \frac{u}{u^2 + v^2} \quad \text{and} \quad y = \frac{-v}{u^2 + v^2} \quad (2) \quad (6)$$

$$x^2 + y^2 - 6x = 0$$

$$\left(\frac{u}{u^2 + v^2}\right)^2 + \left(\frac{-v}{u^2 + v^2}\right)^2 - 6u = 0 \quad (2)$$





After simplification

$$u = \frac{1}{6}, \text{ which is a straight line in the } W \text{ plane.} \quad (2)$$

#### Module 4

$$17(a) \oint_C \frac{f(z)dz}{(z-z_0)^4} dz = \frac{2\pi i}{3!} f^3(z_0) . \quad (1) \quad (7)$$

$$\oint_C \frac{\sin 2z}{(z-0)^4} = \frac{2\pi i}{3!} f^3(0) . \quad (2)$$

Where  $f(z) = \sin 2z$ .

Finding  $f^3(0)$  and substituting in the formula (3)

$$\oint_C \frac{\sin 2z dz}{(z-0)^4} = \frac{-8}{3} \pi i \quad (1)$$

$$(b) \quad f(z) = \frac{1}{1+z} = \frac{1}{1+z+i-i} \quad (1) \quad (7)$$

$$= \frac{1}{1+i} \left(1 + \left(\frac{z-i}{1+i}\right)\right)^{-1} \quad (3)$$

$$= \frac{1}{1+i} - \frac{(z-i)}{(1+i)^2} + \frac{(z-i)^2}{(1+i)^3} + \dots \quad (3)$$

$$18(a) \int_0^{1+i} (x - y + ix^2) dz \quad (7)$$

$$y = x^2$$

$$dy = 2x dx.$$

$$z = x + iy$$

$$dz = dx + i dy$$

$$dz = dx + i 2x dx = (1 + 2ix) dx \quad (2)$$

$$\int_0^{1+i} (x - y + ix^2) (1 + 2ix) dx = \int_0^{1+i} (x - x^2 + ix^2) (1 + 2ix) dx$$

$$(3) \quad = \frac{-1}{3} + \frac{i}{2} \quad (2)$$

$$(b) \quad (7)$$

The function  $\log z$  is analytic at all points inside and on the circle

$$|z - 3| = 2. \quad (1).$$



The singularity  $z=4$  lie inside the circle  $|z-3|=2$ . (1)

Applying Cauchy's integral formula (4)

$$\oint_C \frac{\log z}{(z-4)^2} dz = \frac{\pi i}{2} \quad (1)$$

### Module 5

$$19(a) \quad f(z) = \frac{z^2-1}{z^2+5z+6} = 1 - \left( \frac{-3}{z+2} + \frac{8}{z+3} \right) \quad (1) \quad (5)$$

$$= 1 - \left( \frac{-3}{z} \left(1 + \frac{2}{z}\right)^{-1} + \frac{8}{3} \left(1 + \frac{z}{3}\right)^{-1} \right). \quad (2)$$

$$= 1 - \left( \frac{8}{3} - \frac{8z}{9} + \frac{8z^2}{27} + \dots + \frac{-3}{z} + \frac{6}{z^2} - \frac{12}{z^3} \dots \right) \quad (2)$$

$$(b) \quad \int_0^{2\pi} \frac{d\theta}{\sqrt{2}-\cos\theta} \quad (9)$$

Consider the contour  $C: |z|=1$ . (1).

$$Z = e^{i\theta} \quad d\theta = \frac{dz}{iz}$$

$$\cos \theta = \frac{z^2+1}{2z}$$

$$I = \int_0^{2\pi} \frac{d\theta}{\sqrt{2}-\cos\theta} = 2i \int_{|z|=1} \frac{dz}{z^2-2\sqrt{2}z+1} \quad (2).$$

$$\text{Poles } z = \sqrt{2} + 1, \sqrt{2} - 1. \quad (1).$$

$$\sqrt{2} - 1 \text{ lie inside the region} \quad (1)$$

$$\text{Residue } \frac{-1}{2}. \quad (2).$$

Applying Residue theorem and finding the answer

$$I = 2\pi. \quad (2)$$

$$20(a) \quad \oint_C \frac{z-23}{z^2-4z-5} dz \text{ where } C: |z-2-i|=3.2. \quad (5)$$

$$\text{Singular points } z = 5, -1. \quad (1)$$

$$\text{Residue of } f(z) \text{ at } z=5 \text{ is } -3. \quad (1)$$

$$\text{Residue of } f(z) \text{ at } z=-1 \text{ is } 4. \quad (1)$$

$$\oint_C \frac{z-23}{z^2-4z-5} dz = 2\pi i. \quad (2)$$



(b)

(9)

$$\int_0^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)}. \quad (1)$$

Consider the contour integral

$I = \int_C f(z)dz$  where  $f(z) = \frac{(z^2 + 2)dz}{(z^2 + 1)(z^2 + 4)}$ . where  $C$  is the path consisting of the semi circular path  $C_R$  along with the diameter from  $-R$  to  $+R$ .

(1).

Poles are  $z = \pm i$  and  $z = \pm 2i$ .

(1)

$z = i$  and  $z = 2i$  lie inside the region.

(1)

Residue of  $f(z)$  at  $z = i$  is  $\frac{1}{6i}$

(1)

Residue of  $f(z)$  at  $z = 2i$  is  $\frac{1}{6i}$ .

(1)

Applying Residue theorem and finding the value of the integral

$$\text{as } \int_0^{\infty} \frac{(x^2 + 2)dx}{(x^2 + 1)(x^2 + 4)} = \frac{\pi}{3}. \quad (3).$$

NB: Any alternative correct answers will get full marks.

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech S1,S2 (S) Examination September 2020 (2015 Scheme)

**Course Code: MA102****Course Name: DIFFERENTIAL EQUATIONS**

Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions, each carries 3 marks*

- 1 Find the ODE  $y'' + ay' + by = 0$  for the basis  $\{e^x, xe^x\}$  (3)
- 2 Reduce to first order and solve  $2xy'' = 3y'$ . (3)
- 3 Find the particular integral of  $y'' + 4y' + 4y = x^2$ . (3)
- 4 Using a suitable transformation, convert the differential equation  $(x^2D^2 - 4xD + 6)y = x$  into a linear differential equation with constant coefficients. (3)
- 5 If  $f(x)$  is a periodic function of period  $2L$  defined in  $[-L, L]$ . Write down Euler's Formulas  $a_0, a_n, b_n$  for  $f(x)$ . (3)
- 6 Find the Fourier series of the function  $f(x) = x$  in the range  $-\pi < x < \pi$ . (3)
- 7 Find the PDE by eliminating arbitrary constants  $a$  and  $b$  from  $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ . (3)
- 8 Find the particular integral of  $(D^3 - 4D^2D' + 4DD'^2)z = \cos(2x + y)$ . (3)
- 9 Write all possible solutions of one dimensional wave equation. (3)
- 10 A homogeneous string is stretched and its ends are fixed at  $x = 0$  and  $x = 40$ . Motion is started by displacing the string into the form  $f(x) = \sin\left(\frac{\pi x}{40}\right)$  from which it is released at time  $t = 0$ . Write the boundary and initial conditions.. (3)
- 11 Solve one dimensional heat equation for  $\lambda < 0$ . (3)
- 12 Find the steady state temperature distribution in a rod of length 40 cm if the ends are kept at  $0^\circ\text{C}$  and  $100^\circ\text{C}$ . (3)

**PART B***Answer six questions, one full question from each module***Module 1**

13 a) Solve  $y'' - 2y' - 3y = 0, y(-1) = e, y'(1) = -\frac{e}{4}$ . (6)

b) Show that the functions  $x^3$  and  $x^5$  are the basis of solutions of ODE  
 $x^2y'' - 7xy' + 15y = 0$ . (5)

**OR**

14 a) Solve ODE  $y^v - 3y^{iv} + 3y''' - y'' = 0$ . (6)

b) Solve the ODE  $xy'' + 2y' + xy = 0$ . Given that  $y_1 = \frac{\cos x}{x}$  is a solution. (5)

**Module 1I**

15 a) By the method of variation of parameters, solve  $y'' + y = \sec x$ . (6)

b) Solve  $x^2y'' - 4xy' + 6y = x^5$ . (5)

**OR**

16 a) Solve  $(2x+3)^2y'' - 2(2x+3)y' - 12y = 6x$ . (6)

b) Solve  $y'' + 2y' - 3y = e^{2x} \sin x$ . (5)

**Module 1II**

17 a) Find the Fourier series of  $f$  defined by  $f(x) = e^x$  in  $(-\pi, \pi)$ . (11)

**OR**

18 a) Obtain Fourier series for the function  $f(x) = x^2, -\pi \leq x \leq \pi$ . (6)

b) Expand  $f(x) = \cos x$  as a half range sine-series in  $0 \leq x \leq \pi$ . (5)

**Module 1V**

19 a) Solve  $r + s - 2t = \sqrt{2x + y}$ . (6)

b) Find the general solution of  $x^2p + y^2q = (x + y)z$ . (5)

**OR**

20 a) Solve  $4r + 12s + 9t = e^{3x-2y}$ . (6)

b) Solve  $(D^2 - DD' - 6D'^2)z = xy$ . (5)

**Module V**

21 a) Using method of separation of variables, solve  $y^2u_x - x^2u_y = 0$ . (5)

b) Find the displacement of a finite string of length  $l$  that is fixed at both ends and is released from rest with an initial displacement of  $2 \sin\left(\frac{\pi x}{l}\right)$ . (5)

OR

- 22 Derive one dimensional wave equation. (10)

**Module VI**

- 23 A rod of length  $L$  is heated so that its ends A and B are at zero temperature .If its initial temperature is given by  $u = \frac{cx(L-x)}{L^2}$ , find the temperature at time  $t$ . (10)

OR

- 24 A rod of length 40cm has its ends A and B kept at  $0^{\circ}\text{C}$  and  $100^{\circ}\text{C}$  respectively until steady state conditions prevail. Suddenly the temperature at A is raised to  $20^{\circ}\text{C}$  and the end B is decreased to  $60^{\circ}\text{C}$  . Find the temperature distribution in the rod at time  $t$ . (10)

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## APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

**Scheme of Valuation/Answer Key**      Total pages: 04  
(Scheme of evaluation (marks in brackets) and answers of problems/key)

**B.TECH (2015 SCHEME) S1S2 (S) EXAMINATION, SEPTEMBER 2020**

**Course Code: MA102**

**Course Name: DIFFERENTIAL EQUATIONS**

Max. Marks: 100

Duration: 3 Hours

### PART A

*Answer all questions, each carries 3 marks*

- 1      1 is a double root ,  $m^2 - 2m + 1 = 0 \rightarrow (2)$ ,  $y'' - 2y' + y = 0 \rightarrow (1)$  (3)
- 2       $z = y'$ ,  $z' = y'' \rightarrow (1)$ ,  $z = cx^{\frac{3}{2}} \rightarrow (1)$ ,  $y = c \frac{2}{5} x^{\frac{5}{2}} + d \rightarrow (1)$  (3)
- 3       $PI = \frac{x^2}{(D+2)^2} \rightarrow (2)$ ,  $= \frac{1}{4} \left(1 + \frac{D}{2}\right)^{-2} (x^2)$ ,  $= \frac{1}{4} \left(x^2 - 2x + \frac{3}{2}\right) \rightarrow (1)$  (3)
- 4       $z = e^x$ ,  $D_1 = \frac{d}{dz} \rightarrow (1)$ ,  $(D_1^2 - 5D_1 + 6)y = \ln z \rightarrow (2)$  (3)
- 5       $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx \rightarrow (1)$ ,  $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \rightarrow (1)$ , (3)  
 $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \rightarrow (1)$
- 6       $f(x)$  is odd , or  $a_0 = a_n = 0 \rightarrow (2)$ ,  $b_n = \frac{2(-1)^{n+1}}{n} \rightarrow (1)$ ,  $f(x) =$  (3)  
 $\sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nx$
- 7       $a^2 = \frac{x}{p} \rightarrow (1)$ ,  $b^2 = \frac{y}{p} \rightarrow (1)$ ,  $2z = px + qy \rightarrow (1)$  (3)
- 8       $PI = \frac{1}{D^3 - 4D^2D' + 4DD'^2} \cos(2x + y) \rightarrow (1)$  (3)  
 $\dots \dots = \frac{1}{D^2 - 4DD' + 4D'^2} \frac{1}{2} \sin(2x + y) \rightarrow (1)$   
 $\dots \dots = \frac{x^2}{2} \frac{1}{2} \sin(2x + y) \rightarrow (1)$
- 9       $u = (c_1x + c_2)(c_3t + c_4) \rightarrow (1)$ ,  $u = (c_1e^{px} + c_2e^{-px})(c_3e^{cpt} + c_4e^{-cpt}) \rightarrow (1)$ , (3)  
 $(c_1 \cos px + c_2 \sin px)(c_3 \cos cpt + c_4 \sin cpt) \rightarrow (1)$



- 10  $y(0, t) = 0, y(40, t) = 0, \frac{\partial y}{\partial t} = 0$  when  $t = 0, y(x, 0) = \sin \frac{\pi x}{40}$  (Any 3 conditions 3 marks) (3)

11  $\lambda = -p^2 \rightarrow (1), X = c_1 \cos px + c_2 \sin px \rightarrow (1), T = c_3 e^{-hp^2 t} \rightarrow (1)$  (3)

12  $u(0, t) = 0 \rightarrow (1), u(40, t) = 100 \rightarrow (1), u = \frac{100}{40} x \rightarrow (1)$  (3)

### PART B

*Answer six questions, one full question from each module*

#### Module 1

13 a) Auxiliary Equation  $m^2 - 2m - 3 = 0 \rightarrow (2), m = 3, m = -1 \rightarrow (2),$   
 $y = c_1 e^{3x} + c_2 e^{-x} \rightarrow (2), c_1 = 10.23, c_2 = \frac{13}{16},$   
 $y = 10.23 e^{3x} + \frac{13}{16} e^{-x}$  (6)

b) Showing  $y_1 = x^3$  and  $y_2 = x^5$  is solution  $\rightarrow (4), \frac{y_1}{y_2} = \frac{1}{x^2} \neq \text{constant or } W \neq 0 \rightarrow$   
 $(1), y_1 = x^3$  and  $y_2 = x^5$  form basis (5)

### OR

14 a) Auxiliary equation  $m^2(m-1)^3 = 0 \rightarrow (2), m_1 = m_2 = 0 \rightarrow (1),$   
 $m_3 = m_4 = m_5 = 1 \rightarrow (1)$   
 $y = c_1 + c_2 x + (c_3 + c_4 x + c_5 x^2) e^x \rightarrow (2)$  (6)

b)  $y'' + \frac{2}{x} y' + y = 0$  (standard form)  $\rightarrow (1), y_2 = u y_1 \rightarrow (1)$   
 $u = \int \frac{1}{y_1^2} e^{\int -p dx} dx \rightarrow (1), y_2 = \frac{\sin x}{x} \rightarrow (2)$  (5)

#### Module 1I

15 a)  $m^2 + 1 = 0 \rightarrow (1), CF = c_1 \cos x + c_2 \sin x \rightarrow (1), W = 1 \rightarrow (2)$   
solving,  $PI = \cos x \log |\cos x| + x \sin x \rightarrow (2)$   
 $y = CF + PI$

b) Putting  $x = e^z \rightarrow (1),$  converting to  $(D_1^2 - 5D_1 + 6)y = e^{5z} \rightarrow (1),$   
 $CF = c_1 x^2 + c_2 x^3 \rightarrow (1), PI = \frac{1}{6} x^5 \rightarrow (1), y = CF + PI \rightarrow (1)$  (5)

### OR

16 a) Put  $2x + 3 = e^z \rightarrow (1),$  converting to  $(D_1^2 - 2D_1 - 3)y = \frac{3}{4} e^z - \frac{9}{4} \rightarrow (2),$  (6)





$$CF = \frac{c_1}{2x+3} + c_2(2x+3)^3 \rightarrow (1), \text{ finding } PI = \frac{3}{16} - \frac{3x}{8} \rightarrow (2)$$

$$b) \quad m^2 + 2m - 3 = 0, m = -3, 1 \rightarrow (1), \quad (5)$$

$$CF = c_1 e^{-3x} + c_2 e^x \rightarrow (1),$$

$$PI = (e^{2x}) \frac{1}{(D+2)^2 + 2(D+2) - 3} (\sin x) \rightarrow (1)$$

$$(e^{2x}) \frac{1}{6D+4} (\sin x) \quad (1) \quad \text{Solving..... } PI = \frac{e^{2x}}{26} (2\sin x - 3\cos x) \rightarrow (1)$$

### Module 1II

$$17 \quad a) \quad a_0 = \frac{2\sinh\pi}{\pi} \rightarrow (3), a_n = 2(-1)^n \frac{\sinh\pi}{\pi(1+n^2)} \rightarrow (4), \quad (11)$$

$$b_n = -\frac{2n(-1)^n \sinh\pi}{\pi(1+n^2)} \rightarrow (4)$$

### OR

$$18 \quad a) \quad a_0 = \frac{2\pi^2}{3} \rightarrow (2), b_n = 0 \text{ or even function} \rightarrow (2), a_n = \frac{4}{n^2} (-1)^n \rightarrow (2) \quad (6)$$

$$b) \quad f(x) = \sum f(x) \sin nx \quad (1) \quad b_n = \frac{2}{\pi} \int_0^\pi f(x) \sin nx \, dx \quad (1) \quad (5)$$

$$b_n = \frac{2n(1+(-1)^n)}{\pi(n^2-1)}, n \neq 1 \quad (2) \quad b_1 = 0 \quad (1)$$

### Module 1V

$$19 \quad a) \quad \text{A.E} \quad m^2 + m - 2 = 0 \dots \dots (1) \quad m = 1, -2 \rightarrow (2), CF = f(y+x) + g(y-2x) \dots \dots (2), P.I = \frac{1}{15} (2x+y)^{\frac{5}{2}} \rightarrow (1) \quad (6)$$

$$b) \quad \frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{(x+y)z} \rightarrow (1), \frac{1}{x} - \frac{1}{y} = c_1 \rightarrow (2), \frac{x-y}{z} = c_2 \text{ or } \frac{xy}{z} \rightarrow (2) \quad (5)$$

### OR

$$20 \quad a) \quad \text{AE} \quad 4m^2 + 12m + 9 = 0 \dots (1) \quad m = -\frac{3}{2}, -\frac{3}{2} \rightarrow (2), \quad (6)$$

$$CF \dots \dots (2), \quad PI = \frac{x^2}{8} e^{3x-2y} \rightarrow (1)$$

$$b) \quad \text{AE} \quad (1) \quad m = 3, -2 \rightarrow (2), CF = f(y+3x) + g(y-x) \rightarrow (1), PI = \frac{x^3}{6} y + \frac{x^4}{24} \rightarrow (1), \quad (5)$$

### Module V

$$21 \quad a) \quad \text{Put } u = XY \rightarrow (1), X = c_1 e^{\frac{kx^3}{3}} \rightarrow (2), Y = c_2 e^{\frac{ky^3}{3}} \rightarrow (2) \quad (5)$$

$$b) \quad \text{Initial and boundary conditions} \rightarrow (2), c_1 = 0, p = \frac{n\pi}{l} \rightarrow (1), a_1 = 2, a_n = \quad (5)$$



0 for  $n \neq 1 \rightarrow (2)$

**OR**

22 Assumptions...(2) Rest.....(6).....Wave Equation....(2)

**Module VI**

23 Solution of Heat equation  $\rightarrow (2)$ , *boundary and initial conditions*  $\rightarrow (2)$ ,

$$u(0, t) = Ae^{\frac{\lambda^2 t}{h}}, A = 0 \rightarrow (2), \quad u(x, t) = B \sin \lambda L e^{\frac{\lambda^2 t}{h}} = 0 \rightarrow \lambda = \frac{n\pi}{L} \rightarrow (2), \quad (10)$$

$$B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{l} dx = \frac{4c}{n^3 \pi^3} (1 - \cos n\pi) \quad (2)$$

**OR**

24 *steady state*,  $u = \frac{5}{2}x \rightarrow (3)$ , *initial and boundary conditions*  $\rightarrow (1)$ , (10)

$$u(x, t) = u_1(x) + u_2(x, t) \rightarrow (1), u_1(x) = x + 20 \rightarrow (2)$$

$$u_2(x, 0) = \frac{60}{40}x - 20 \rightarrow (1)$$

$$B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{l} dx = -\frac{40}{n\pi} (1 + 2(-1)^n) \rightarrow (2),$$

\*\*\*\*

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**  
Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

**Course Code: MAT201**

**Course Name: Partial Differential equations and Complex analysis**

Max. Marks: 100

Duration: 3 Hours

**PART A**

*Answer all questions. Each question carries 3 marks*

Marks

- 1 Form the PDE for the equation  $z = f(x^2 - y^2)$  where  $f$  is an arbitrary function. (3)
- 2 Solve  $\frac{\partial^2 z}{\partial x^2} + z = 0$  given that when  $x = 0, z = e^y$  and  $\frac{\partial z}{\partial x} = 1$  (3)
- 3 Write the conditions in which a tightly stretched string of length  $l$  with fixed end is initially in equilibrium position and is set vibrating by giving each point a velocity  $v_0 \sin^3 \frac{\pi x}{l}$  (3)
- 4 Write down the possible solutions of one dimensional heat equation. (3)
- 5 Find the real part and imaginary part of the function  $f(z) = 5z^2 - 12z + 3 + 2i$  and find their values at  $z = 4 - 3i$  (3)
- 6 Find the fixed points of the mapping  $w = (a + ib)z^2$  (3)
- 7 Evaluate  $\oint_C \frac{e^z}{z-2} dz$  where  $C$  is  $|z| = 3$  (3)
- 8 Find the Maclaurin series expansion of  $\int_0^z e^{-t^2} dt$  (3)
- 9 Find the Laurent series of  $z^{-5} \sin z$  with centre 0 (3)
- 10 Find the residue at poles for the function  $f(z) = \frac{-8}{1+z^2}$  (3)

**PART B**

*Answer any one full question from each module. Each question carries 14 marks*

**Module 1**

- 11(a) Solve  $(x^2 - y^2 - z^2)p + 2xyq = 2xz$  (7)
- (b) Solve  $(p^2 + q^2)y = qz$  by Charpit's method (7)
- 12(a) Find the differential equation of all planes which are at a constant distance 'a' from the origin (7)
- (b) Solve  $3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$  where  $u(x, 0) = 4e^{-x}$  by the method of separation of variables. (7)

**Module 2**

- 13 The points of trisection of a string are pulled aside through the same distance on opposite sides of the position of equilibrium and the string is released from rest. Derive an expression for the displacement of the string at subsequent time. (14)
- 14 Solve the boundary value problem  $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ ,  $0 < x < l$ ,  $\frac{\partial u}{\partial x}(0, t) = 0$ ,  $\frac{\partial u}{\partial x}(l, t) = 0$ ,  $u(x, 0) = x$  (14)

**Module 3**

- 15(a) Determine 'a' so that the function  $u = e^{-\pi x} \cos y$  is harmonic and find its harmonic conjugate. (7)
- (b) Is the function  $f(z) = \begin{cases} \frac{\operatorname{Re} z^2}{|z|^2}, & z \neq 0 \\ 0, & z = 0 \end{cases}$  continuous or not at  $z = 0$  (7)
- 16(a) Find the image of the region  $|z - \frac{1}{2}| \leq \frac{1}{2}$  under the transformation  $w = \frac{1}{z}$  (7)
- (b) Show that  $f(z) = |z|^2$  is differentiable only at  $z = 0$ , hence it is nowhere analytic. (7)

**Module 4**

- 17(a) Evaluate  $\oint_C \frac{z^3 - 6}{2z - i} dz$  where  $C$  is  $|z| = 1$  in counterclockwise direction (7)
- (b) Find the Maclaurin series expansion of  $\frac{z+2}{1-z^2}$  (7)
- 18(a) Integrate  $\oint_C \frac{\sinh 2z}{(z - \frac{1}{2})^4} dz$  in counterclockwise direction around the unit circle (7)
- (b) Find the Taylor series expansion of  $f(z) = \frac{1}{1+z}$  with centre  $z_0 = -i$  (7)

**Module 5**

- 19(a) Find the Laurents series of that  $f(z) = \frac{e^z}{(z-1)^2}$  that converge for  $0 < |z - 1| < R$  and determine the region of convergence (7)
- (b) Evaluate  $\oint_0^{2\pi} \frac{\sin^2 \theta}{5 - 4 \cos \theta} d\theta$  (7)
- 20(a) Evaluate  $\oint_C \tan 2\pi z dz$  counter clockwise around  $C : |z - 0.2| = 0.2$  (7)
- (b) Find the principal value of  $\int_{-\infty}^{\infty} \frac{dx}{(x^2 - 3x + 2)(x^2 + 1)}$  (7)

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## APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

### Scheme for Valuation/Answer Key

*Scheme of evaluation (marks in brackets) and answers of problems/key*  
**Third Semester B.Tech Degree Examination December 2020 (2019 scheme)**

**Course Code: MAT201**

**Course Name: Partial Differential equations and Complex analysis**

Max. Marks: 100

Duration: 3 Hours

#### PART A

*Answer all questions. Each question carries 3 marks*

- |   |                                                                                                                                                                                              | Marks |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | $p = f'(x^2 - y^2)2x$ (1 mark)<br>$q = f'(x^2 - y^2)(-2y)$ (1 mark)<br>$py + qx = 0$ is the p d e (1 mark)                                                                                   | (3)   |
| 2 | $z = f(y)\sin x + \phi(y)\cos x$ (1 mark)<br>Applying conditions $f(y) = 1, \phi(y) = e^y$ (2 marks)                                                                                         | (3)   |
| 3 | $y(0, t) = 0, y(l, t) = 0, y(x, 0) = 0, \frac{\partial y}{\partial t}(x, 0) = v_0 \sin^3 \frac{\pi x}{l}$<br>Write any 3 conditions (give 1 mark to each ).                                  | (3)   |
| 4 | $u(x, t) = (Ae^{px} + Be^{-px})e^{c^2 p^2 t}$ (1 mark)<br>$u(x, t) = (A \cos px + B \sin px)e^{-c^2 p^2 t}$ (1 mark)<br>$u(x, t) = (Ax + B)$ (1 mark)                                        | (3)   |
| 5 | Substitute $z = x + iy$ in $f(z)$<br>Separating real and imaginary parts, $u = 5x^2 - 5y^2 - 12x + 3$ (1 mark)<br>$v = 10xy - 12y + 2$ (1 mark)<br>$u(4, -3) = -10, v(4, -3) = -82$ (1 mark) | (3)   |
| 6 | $f(z) = z \Rightarrow (a + ib)z^2 = z$ ( 2 marks )                                                                                                                                           | (3)   |



- Solving,  $z = 0, \frac{a-ib}{a^2+b^2}$  or  $\frac{1}{a+ib}$  (1 marks)
- 7 Singular point  $z=2$  is inside  $|z| = 3$  ( 2 marks ) (3)
- By Cauchy's integral formula ,  $\oint_C \frac{e^z}{z-2} dz = 2\pi i e^2$  (1 mark)
- 8  $\int_0^z e^{-t^2} dt = \int_0^z 1 - \frac{t^2}{2} + \frac{t^4}{8} - \dots dt$  (2 marks) (3)
- $= z - \frac{z^3}{6} + \frac{z^5}{40} + \dots$  .(1 mark )
- 9  $z^{-5} \sin z = \frac{\sin z}{z^5}$  (1 mark)  $= \frac{1}{z^5} [z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots]$  (2 marks) (3)
- 10  $z = \pm i$  are simple poles (1 mark) (3)

$$\text{Res}_{z=i} f(z) = 4i \text{ (1 mark)}$$

$$\text{Res}_{z=-i} f(z) = -4i \text{ (1 mark)}$$

## PART B

Answer any one full question from each module. Each question carries 14 marks

### Module 1

11(a) (7)

The auxiliary equation is  
 $\frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$  ( 2 marks )

Taking last two ratios:  $\frac{dy}{2xy} = \frac{dz}{2xz}$

$$\Rightarrow \frac{dy}{y} = \frac{dz}{z}$$

Integrating  $\log y = \log z + \log a$

$$\Rightarrow y/z = a \text{ ( 2 marks )}$$

Taking Lagrangian multipliers as,  $x, y, z$  each ratios of (1 mark)

$$= \frac{xdx+yd y+zd}{x(x^2+y^2+z^2)}$$

$$\text{Now take , } \frac{dy}{2xy} = \frac{xdx+yd y+zd}{x(x^2+y^2+z^2)} \text{ ( 1 mark )}$$



$$\Rightarrow \frac{dy}{y} = \frac{d(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2}$$

Integrating

$$\log y = \log(x^2 + y^2 + z^2) + \log b$$

$$\frac{y}{x^2 + y^2 + z^2} = b \quad (1 \text{ mark})$$

The general equation is

$$\phi\left(\frac{y}{z}, \frac{y}{x^2 + y^2 + z^2}\right) = 0$$

(b) Let  $f(x, y, z, p, q) = (p^2 + q^2)y - qz = 0$  (1 mark)

(7)

Charpit's subsidiary equation is  $\frac{dx}{-2py} = \frac{dy}{z-2qy} = \frac{dz}{-q} = \frac{dp}{-pq} = \frac{dq}{p^2}$  (2 marks)

$$\frac{dp}{-pq} = \frac{dq}{p^2}$$

Integrating,  $p^2 + q^2 = a$  (1 mark)

Solving and substituting in  $z^2 - pqxy = 0$ ,  $q = \frac{z}{ay}$  (1 mark)

From  $(p^2 + q^2)y = qz$ ,  $q = \frac{c^2 y}{z}$ ,  $p = \frac{c\sqrt{z^2 - c^2 y^2}}{z}$  (1 mark)

$$dz = p dx + q dy$$

Solving,  $z^2 = (a + cx)^2 + c^2 y^2$  (1 mark)

12(a) Equation of plane is  $lx + my + nz = a$  where  $l^2 + m^2 + n^2 = 1$  (7)  
2 marks)

$$lx + my + \sqrt{1 - l^2 - m^2} z = a\sqrt{l^2 + m^2 + n^2} \quad (2 \text{ marks})$$

Differentiating partially w.r.to x and y,  $l = -np$  (1 mark),  $m = -nq$  (1 mark)



Substituting in  $lx + my + \sqrt{1 - l^2 - m^2}z = a\sqrt{l^2 + m^2 + n^2}$   
and solving

$$z = px + qy + a\sqrt{p^2 + q^2 + 1} \quad (1 \text{ mark})$$

b)

Solution is  $u(x, t) = X(x)T(t)$  (2 mark)

Substituting in Pde and simplifying  $\frac{3X'}{X} = \frac{-2Y'}{Y} =$

$k, a \text{ constant}$  (2 mark)

$$\frac{3X'}{X} = k$$

Integrating  $X = C_1 e^{\frac{kx}{3}}$  (1 mark)

$$\frac{-2Y'}{Y} = k \Rightarrow Y = C_2 e^{-\frac{k}{2}} \quad (1 \text{ mark})$$

$$u = C_1 C_2 e^{\frac{kx}{3}} e^{-\frac{k}{2}}$$

$$u(x, 0) = 4e^{-x} \Rightarrow C_1 C_2 = 4, \quad k = -3$$

$$u = 4e^{-x + \frac{3y}{2}} \quad (1 \text{ mark})$$

### Module 2

13

The equation of the vibrating string is  $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$  (1 mark) (14)

Boundary conditions- 1)  $y(0, t) = 0$ , 2)  $y(l, t) = 0$  (2 marks)

Initial conditions - 3)  $\frac{\partial y}{\partial t}_{t=0} = 0$ , (1 mark)

$$y(x, 0) = \begin{cases} \frac{3ax}{l}, & 0 < x < \frac{l}{3} \\ \frac{3a(l-2x)}{l}, & \frac{l}{3} \leq x \leq \frac{2l}{3} \\ \frac{3a(x-l)}{l}, & \frac{2l}{3} \leq x \leq l \end{cases} \quad (3 \text{ marks})$$

$$y = (A \cos px + B \sin px)(C \cos pct + D \sin pct) \quad (3 \text{ marks})$$

(Give 3 marks without derivation)\*\*

By condition 1  $A = 0$  (1 mark)

By condition 2,  $p = \frac{n\pi}{l}$  (1 mark)

By condition (3)  $D=0$  (1 mark)





$$y(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l}; \text{BC} = b_n \text{ (1 mark)}$$

$$\text{By condition 3, } y(x, 0) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \text{ (1 mark)}$$

$$b_n = \frac{2}{l} \left[ \int_0^{\frac{l}{3}} \frac{3ax}{l} \sin \frac{n\pi x}{l} dx + \int_{\frac{l}{3}}^{\frac{2l}{3}} \frac{3a(l-2x)}{l} \sin \frac{n\pi x}{l} dx + \int_{\frac{2l}{3}}^l \frac{3a(x-l)}{l} \sin \frac{n\pi x}{l} dx \right] = \frac{18a}{n^2\pi^2} \sin \frac{n\pi}{3} (1 + (-1)^n) \text{ (1 marks)}$$

$$y(x, t) = \sum_{n=1}^{\infty} \frac{18a}{n^2\pi^2} \sin \frac{n\pi}{3} (1 + (-1)^n) \sin \frac{n\pi x}{l}$$

14

The temperature  $u(x, t)$  is the solution of heat equation  $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$  (14)

(1 mark)

Conditions

$$u(x, t) = (A \cos px + B \sin px) e^{-c^2 p^2 t} \text{ (1 mark)}$$

$$\frac{\partial u}{\partial x} = (-C_1 p \sin px + C_2 p \cos px) e^{-c^2 p^2 t} \text{ (1 mark)}$$

$$\frac{\partial u}{\partial x}(0, t) = 0 \Rightarrow C_2 = 0 \text{ (1 mark)}$$

$$\frac{\partial u}{\partial x}(l, t) = 0 \Rightarrow p = \frac{n\pi}{l} \text{ (1 mark)}$$

$$u(x, t) = C_1 \cos \frac{n\pi x}{l} e^{\frac{-c^2 n^2 \pi^2 t}{l^2}} \text{ (1 mark)}$$

$$u(x, t) = \sum_{n=0}^{\infty} b_n \cos \frac{n\pi x}{l} e^{\frac{-c^2 n^2 \pi^2 t}{l^2}} \text{ (1 mark)}$$

$$u(x, 0) = \sum_{n=0}^{\infty} b_n \cos \frac{n\pi x}{l} = x \text{ (1 mark)}$$



$$b_o = \frac{2}{l} \left[ \int_0^l x \, dx \right] = l (2 \text{ marks})$$

$$b_n = \frac{2}{l} \left[ \int_0^l x \cos \frac{n\pi x}{l} dx \right] = \frac{2l}{n^2\pi^2} [\cos n\pi - 1] (3 \text{ marks})$$

$$u(x, t) = \frac{l}{2} + \sum_{n=0}^{\infty} \frac{2l}{n^2\pi^2} [\cos n\pi - 1] \cos \frac{n\pi x}{l} e^{-\frac{c^2 n^2 \pi^2 t}{l^2}} (1 \text{ mark})$$

**\*\* (give 10 marks to reasonable attempt )**

### Module 3

$$15(a) \quad u = e^{-\pi x} \cos ay \quad (7)$$

$$u_{xx} = \pi^2 e^{-\pi x} \cos ay \quad (1 \text{ mark}), u_{yy} = -a^2 e^{-\pi x} \cos ay \quad (1 \text{ mark})$$

$$u_{xx} + u_{yy} = 0 \quad (1 \text{ mark})$$

$$\text{Substitute and solve, } a = \pm \pi, u = e^{-\pi x} \cos \pi y \quad (1 \text{ mark})$$

$$\text{Using CR equations, } v_y = -\pi e^{-\pi x} \cos \pi y \quad (1 \text{ mark}),$$

$$v_x = \pi e^{-\pi x} \sin \pi y \quad (1 \text{ mark})$$

Integrating any one and comparing with other,  $v = e^{-\pi x} \sin \pi y + c$  (1 marks) **\*\* (Also give marks for other methods )**

$$(b) \quad \lim_{z \rightarrow 0} \operatorname{Re} \frac{z^2}{|z|^2} = \lim_{z \rightarrow 0} \frac{\operatorname{Re}(x+iy)^2}{|x+iy|^2} = \lim_{z \rightarrow 0} \frac{x^2 - y^2}{x^2 + y^2} \quad (4 \text{ marks}) \quad (7)$$

Consider the path  $y = mx$  (1 mark)

$$\lim_{z \rightarrow 0} \operatorname{Re} \frac{z^2}{|z|^2} = \lim_{x \rightarrow 0} \frac{x^2 - m^2 x^2}{x^2 + m^2 x^2} = \frac{1 - m^2}{1 + m^2}, \text{ not unique} \quad (1 \text{ mark})$$

Hence  $f$  is not continuous at  $z=0$  (1 mark)

$$16(a) \quad \left| x + iy - \frac{1}{2} \right| = \frac{1}{2} \quad (1 \text{ marks}) \quad (7)$$

$$x^2 + y^2 - x = 0 \quad (2 \text{ marks})$$

$$w = \frac{1}{z} \Rightarrow x = \frac{u}{u^2 + v^2} \quad (1 \text{ mark}) \quad y = \frac{-v}{u^2 + v^2} \quad (1 \text{ mark})$$

Substituting solving  $u = 1$  (1 marks)



(b) Hence  $\left|z - \frac{1}{2}\right| \leq \frac{1}{2} \Rightarrow u \leq 1$  (1 mark)

$$f(z) = |z|^2 \Rightarrow u = x^2 + y^2, v = 0 \text{ (2 marks)} \quad (7)$$

CR equations – (2 mark)

Verify CR equations (1 mark)

At  $z=0$  only CR equations are satisfied (1 mark)

Partial derivatives - continuous

Conclusion (1 mark)

#### Module 4

17(a) Given function is not analytic at  $z = \frac{i}{2}$  which is inside C (2 marks) (7)

)

Cauchy integral formula (2 marks)

$$f\left(\frac{i}{2}\right) = \frac{i^3}{8} - 6 \text{ (1 mark)}$$

Applying formula (1 marks)

(b) Ans:  $-\pi i \left(\frac{-i}{8} - 6\right) = \frac{\pi}{8} - 6\pi i$  (1 marks) (7)

$$\frac{z+2}{1-z^2} = \frac{z+2}{(1-z)(1+z)} \text{ (2 marks)}$$

Apply partial fractions,  $= \frac{1}{2(1+z)} + \frac{3}{2(1-z)}$  (2 marks)

$$= \frac{1}{2}(1+z)^{-1} + \frac{3}{2}(1-z)^{-2} \text{ (2 marks)}$$

Expanding and solving,  $2+z+2z^2+3z^3+\dots$  (1 mark)

**\*\* (Give marks for other methods )**

G 18(a) Given function is not analytic at  $z = \frac{1}{2}$  which is inside C (2 marks) (7)

)

Cauchy integral formula for derivatives (2 marks)

Applying formula (2 marks)

(b) simplyfing Ans:  $\frac{8\pi i}{3} \cosh 1$  (1 mark) (7)

$$\frac{1}{1+z} = \frac{1}{1+z+i-i} \text{ (2 marks)}$$



$$= \frac{1}{1-i} \left[ \frac{1}{1+\frac{z+i}{1-i}} \right] \text{ (2 marks)}$$

$$= \frac{1}{1-i} \left[ 1 + \frac{z+i}{1-i} \right]^{-1} \text{ (2 marks)}$$

Expanding , Ans:-  $\frac{1}{1-i} \left[ 1 - \frac{z+i}{1-i} + \left( \frac{z+i}{1-i} \right)^2 - \dots \right]$  ( 1 mark )

**\*\* ( Give marks for other methods )**

### Module 5

19(a) 
$$\frac{e^z}{(z-1)^2} = \frac{e^{z-1+1}}{(z-1)^2} = \frac{e^{z-1}e}{(z-1)^2} \text{ (3 marks)} \quad (7)$$

$$= e \frac{e^{z-1}}{(z-1)^2} \text{ (1 mark)}$$

$$= \frac{e}{(z-1)^2} \left[ 1 + \frac{z-1}{1!} + \frac{(z-1)^2}{2!} + \dots \right] \text{ (2 marks)}$$

Region is  $0 < |z-1| < \infty$  ( 1 mark )

(b)

Consider contour  $C : |z| = 1 \Rightarrow z = e^{i\theta}, d\theta = \frac{dz}{iz}$  (1 mark)

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5-4\cos \theta} d\theta = \frac{1}{4i} \int_C \frac{(z^4-2z^2+1) dz}{z^2(2z^2-5z+2)} \text{ (1 marks)} \quad (7)$$

$$= \frac{1}{4i} \int_C f(z) dz$$

Singular points are  $z = \frac{1}{2}, 2, 0$  (1 mark)

$z=0, \frac{1}{2}$  lie inside C

$z=0$  is a pole of order 2 and  $z = \frac{1}{2}$  is a pole of order 1 (1 mark)

$$\text{Res } f(z)_{z=0} = \frac{5}{4} \text{ (1 mark)}$$

$$\text{Res } f(z)_{z=\frac{1}{2}} = \frac{-3}{4} \text{ (1 mark)}$$

Apply Cauchy's residue theorem , answer  $= \frac{\pi}{4}$  (1 mark)

20(a) Singular points are  $z = \frac{2n+1}{4}, n = 0, \pm 1, \pm 2, \dots$  (2 marks) (7)



Among these points  $z = \frac{1}{4}$  lie in C (2 mark)

$$\text{Res } f(z)_{z = \frac{1}{4}} = \frac{-1}{2\pi} \text{ (2 marks)}$$

Apply Cauchy's residue theorem, answer = - i (1 mark)

(b)

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2-3x+2)(x^2+1)} = \int_{-\infty}^{\infty} \frac{dz}{(z^2-3z+2)(z^2+1)} \text{ (2 marks)}$$

Singular points are  $z=1, 2, \pm i$  (2 marks)

Among these  $z = i$  lie in upper half plane. (2 marks)

$$\text{Res } f(z)_{z=1} = \frac{-1}{2}$$

$$\text{Res } f(z)_{z=2} = \frac{1}{5}$$

$$\text{Res } f(z)_{z=i} = \frac{3-i}{20} \text{ (1 mark)}$$

$$\text{Principal value} = 2\pi i \left( \frac{3-i}{20} \right) + \pi i \left( \frac{-1}{2} + \frac{1}{5} \right) = \frac{\pi}{10}$$

**\*\*Give full marks for any reasonable attempt.**



**APJ ABDULKALAM TECHNOLOGICAL UNIVERSITY**  
**THIRD SEMESTER B.TECH. DEGREE EXAMINATION MARCH 2021**

**Course Code: MAT201**

**Course Name: PARTIAL DIFFERENTIAL EQUATIONS AND COMPLEX ANALYSIS**

Max. Marks: 100

Duration : 3hrs

**PART A**

**Answer all questions, each carries 3 marks.**

1. Form the differential equation by eliminating the arbitrary function from  $z=x+y+f(xy)$
2. Solve the differential equation  $(y^2+z^2)p-xyq+xz=0$
3. What are the possible solutions of one dimensional heat equation?
4. Find the steady-state temperature distribution in a rod of length 20cm if the ends of the rod are kept at 10°C and 70 °C.
5. Show that the function  $|z|^2$  is not analytic at any point.
6. If  $f(z)=u+iv$  is analytic, prove that  $u=\text{constant}$  and  $v=\text{constant}$  are families of curves cutting orthogonally.
7. Evaluate  $\int_c \frac{e^{3iz}}{(z+\pi)^3} dz$ , where  $c$  is circle  $|z - \pi| = 3/2$ .
8. Integrate  $\frac{z^2}{z^2-1}$  counter clockwise around the circle  $|z - 1 - i| = \frac{\pi}{2}$ .
9. Compute the residue of the function  $f(z) = \frac{4-3z}{z(z-1)(z-2)}$  at each of the poles in  $|z| = 3/2$ .
10. Define three types of isolated singularities with an example for each.

**PART B**

**Answer any one full question from each module, each question carries 14 marks.**

**Module - 1**

11. a) Solve  $y^2p-xyq=x(z-2y)$ . (8)
- b) Form the PDE by eliminating  $a, b, c$  from  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  (6)
12. a) Using method of separation of variables, solve  $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ , where  $u(x, 0) = 6e^{-3x}$  (8)
- b) Solve  $px+qy=pq$ . (6)

**Module - II**

13. a) A tightly stretched string with fixed end points  $x=0$  and  $x=l$  is initially in a position given by  $u(x, 0) = u_0 \sin^3(\frac{\pi x}{l})$ . If it is released from rest from this position, find the displacement  $u$  at any distance  $x$  from one end at any time  $t$ . (8)
- b) Find temperature distribution of a rod of length 2m, whose end points are maintained at temperature zero and the initial temperature is  $f(x)=100(2x-x^2)$ . (6)

14. a) A tightly stretched string with fixed end points  $x=0$  and  $x=l$  is initially at rest in its equilibrium position. If it is set vibrating by giving each of its points a velocity of  $\lambda x(l-x)$ , find the displacement of the string at any distance  $x$  from one end at any time  $t$ . (8)
- b) Find temperature distribution in a metal rod of length  $L$  with insulated ends and with initial temperature in the rod  $\sin\left(\frac{\pi x}{L}\right)$ . (6)

### Module - III

15. a) Find the value of 'a' and 'b' such that the function  $f(z) = x^2 + ay^2 - 2xy + (bx^2 - y^2 + 2xy)$  is analytic. (6)
- b) Show that  $f(z) = e^{-x} \cos y - e^{-x} \sin y$  is differentiable everywhere. Find its derivative. (8)
16. a) Find the image of infinite strip  $\frac{1}{4} < y < \frac{1}{2}$  under the transformation  $w = \frac{1}{z}$ . (6)
- b) Prove that the function  $u(x, y) = x^3 - 3xy^2 - 5y$  is harmonic. Also find harmonic conjugate of  $u$ . (8)

### Module - IV

17. a) Evaluate  $\int_0^{3+i} z^2 dz$  along the parabola  $y^2 = x/3$ . (8)
- b) Find the value of  $\int_c \frac{2z^2 + z}{z^2 - 1} dz$ , where  $c$  is the circle of unit radius with center at  $z=1$ . (6)
18. a) Evaluate  $\int_c \bar{z} dz$  where  $c$  is given by  $x=3t$ ,  $y=t^2$ ,  $-1 \leq t \leq 4$ . (6)
- b) Use Cauchy's integral formula to evaluate  $\int_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$ , where  $c$  is the circle  $|z| = 3$ . (8)

### Module - V

19. a) Expand  $\frac{1}{z^2 - 3z + 2}$  in the region  $|z| < 1$ . (6)
- b) Evaluate  $\int_c \frac{z-1}{(z+1)^2(z-2)} dz$ , where  $c$  is  $|z-2| = 2$  using Cauchy's residue theorem. (8)
20. a) Find the Laurent series expansion of  $\frac{1}{z-z^3}$  in  $1 < |z+1| < 2$ . (6)
- b) Evaluate  $\int_0^{2\pi} \frac{\cos 2\theta}{5 + 4\cos \theta} d\theta$ . (8)